

TRACKING URBAN EXPANSION AND LAND USE CHANGE IN DELHI USING SATELLITE IMAGE PROCESSING

A Minor Project

Submitted in partial fulfilment of the requirement for the award of the degree of

BACHELOR OF TECHNOLOGY IN COMPUTER SCIENCE AND ENGINEERING

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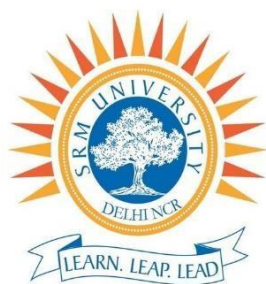
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CANDIDATE'S DECLARATION

We hereby certify that the work which is being presented in the project entitled **“TRACKING URBAN EXPANSION AND LAND USE CHANGE IN DELHI USING SATELLITE IMAGE PROCESSING”** in partial fulfilment of the requirement for the award of the degree of Bachelor of Technology in Computer Science & Engineering and submitted in the Department of Computer Science & Engineering of SRM University, Delhi NCR, Sonapat, Haryana, (India) is an authentic record of our own work carried out under the supervision of **Dr. Ravi Sharma**, as minor project in 7th semester during the academic year 2024-25. The matter presented in this project has not been submitted for the award of any other degree of this or any other Institute/University.

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CERTIFICATE

This is to certify that the project titled " TRACKING URBAN EXPANSION AND LAND USE CHANGE IN DELHI USING SATELLITE IMAGE PROCESSING" is the bonafide work carried out by Apoorva Pahal, Harsh Tyagi and Tanishq Narang, students of B. Tech. (CSE – A&C) of SRM University Delhi-NCR, Sonapat, Haryana-131029 during the academic year 2021-2025, impartial fulfilment of the requirements for the award of the degree of Bachelor of Technology (Computer Science and Engineering) and that the project has formed the basis for the award previously of any other degree, diploma, fellowship or any other similar title.

Dr. Ravi Sharma
(Assistant Professor)

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ABSTRACT

Urban expansion and land use changes significantly impact environmental sustainability, infrastructure planning, and socio-economic dynamics. This project focuses on tracking urban expansion and analyzing land use changes in Delhi, one of the world's fastest-growing metropolises, using satellite image processing techniques. Leveraging multi-temporal remote sensing data from satellites such as Cartosat and Resourcesat, the study employs advanced image processing algorithms, including supervised classification, change detection, and machine learning models. Geographic Information System tools are integrated to map spatial patterns and quantify urban sprawl over time.

The analysis highlights key trends, including the conversion of agricultural and green areas into built-up spaces, and identifies hotspots of rapid urbanization. Results from this study provide critical insights for urban planners, policymakers, and environmentalists to make informed decisions regarding sustainable urban development and resource management. By combining automation in satellite image processing with geospatial analysis, this project sets a foundation for scalable solutions in monitoring urbanization globally.

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INTRODUCTION

GENERAL INTRODUCTION

Satellite image processing has emerged as a crucial technique for monitoring environmental changes, including deforestation, urbanization, and climate impact. The growing availability of high-resolution satellite imagery enables researchers to assess environmental dynamics at a global and regional scale. Forest cover analysis through satellite data provides valuable insights into the extent of deforestation, its drivers, and its impact on biodiversity and ecosystem services. This field plays a pivotal role in supporting international efforts like the Sustainable Development Goals (SDGs), biodiversity conservation, and global climate initiatives such as the Paris Agreement. By leveraging advanced tools and techniques, satellite image processing enables actionable data-driven decisions to mitigate environmental challenges.

PROJECT PURPOSE

The primary objective of this project is to analyze the changes in forest cover in an urban area like Delhi over the past decade. The city of Delhi, with its rapid urbanization and population growth, has witnessed significant environmental changes. By leveraging advanced image processing techniques, this project aims to provide a quantitative and qualitative assessment of deforestation trends. Additionally, this project aims to highlight the relationship between urban expansion and ecological degradation, serving as a basis for recommending sustainable urban development policies, promoting environmental awareness, and guiding future research.

SCOPE:

- Monitoring forest cover changes over a 10-year period using satellite images from multiple sources.
- Implementing advanced image processing techniques, including clustering algorithms and spectral analysis, to ensure accurate results.
- Developing a reproducible workflow using MATLAB that can be applied to similar studies in other regions.
- Enhancing understanding of urbanization's environmental impact to inform policymakers and planners.
- Offering insights into potential reforestation areas and strategies for ecological restoration.
- Exploring the impact of seasonal changes on vegetation health and integrating these findings into predictive models.
- Using historical forest data to compare long-term trends and evaluate policy effectiveness.
- Incorporating socio-economic data to correlate urban expansion patterns with forest loss.

LITERATURE SURVEY

TABULAR REPRESENTATION:

Author(s)	Year	Study Focus	Key Findings
Seto et al.	2011	Urban land expansion and forests.	Highlighted urbanization's impact on forest loss.
Bhatta et al.	2013	Remote sensing for urban analysis.	Demonstrated efficient mapping of urban areas.
Roy et al.	2018	Forest change detection using GIS.	Developed advanced techniques for deforestation monitoring.

PROBLEM STATEMENT

Rapid urbanization in Delhi, driven by population growth and economic development, has resulted in significant deforestation over the past decade. This has led to severe ecological consequences, including habitat loss, increased urban heat islands, and disrupted carbon cycles. The lack of comprehensive, high-resolution, and automated methodologies for monitoring forest changes has made it challenging to quantify the environmental impact accurately. This project addresses these challenges by utilizing satellite imagery, advanced image processing techniques, and spectroscopy-based analysis to deliver accurate and actionable results. The project also seeks to establish a scalable framework that can be applied to other urban areas facing similar challenges.

TECHNOLOGIES USED

- **Satellite Imaging:** ResourceSat and Cartosat satellites provide high-resolution images crucial for detailed analysis.
- **Clustering Techniques:** Algorithms such as K-means and Hierarchical Clustering categorize land cover into vegetation, urban areas, and bare soil.
- **Spectroscopy-based Analysis:** The calculation of NDVI (Normalized Difference Vegetation Index) enables the assessment of vegetation health and coverage.
- **MATLAB:** A versatile computational tool for preprocessing, clustering, spectral analysis, and visualization. MATLAB's advanced libraries facilitate the efficient handling of large datasets and complex calculations.

SOFTWARE REQUIREMENTS

LIBRARIES:

- **MATLAB Image Processing Toolbox:** Used for image enhancement, noise reduction, spatial subset selection, and normalization.
- **MATLAB Statistics and Machine Learning Toolbox:** Essential for implementing clustering algorithms and performing statistical analysis on large datasets.

DESIGN

DESIGN

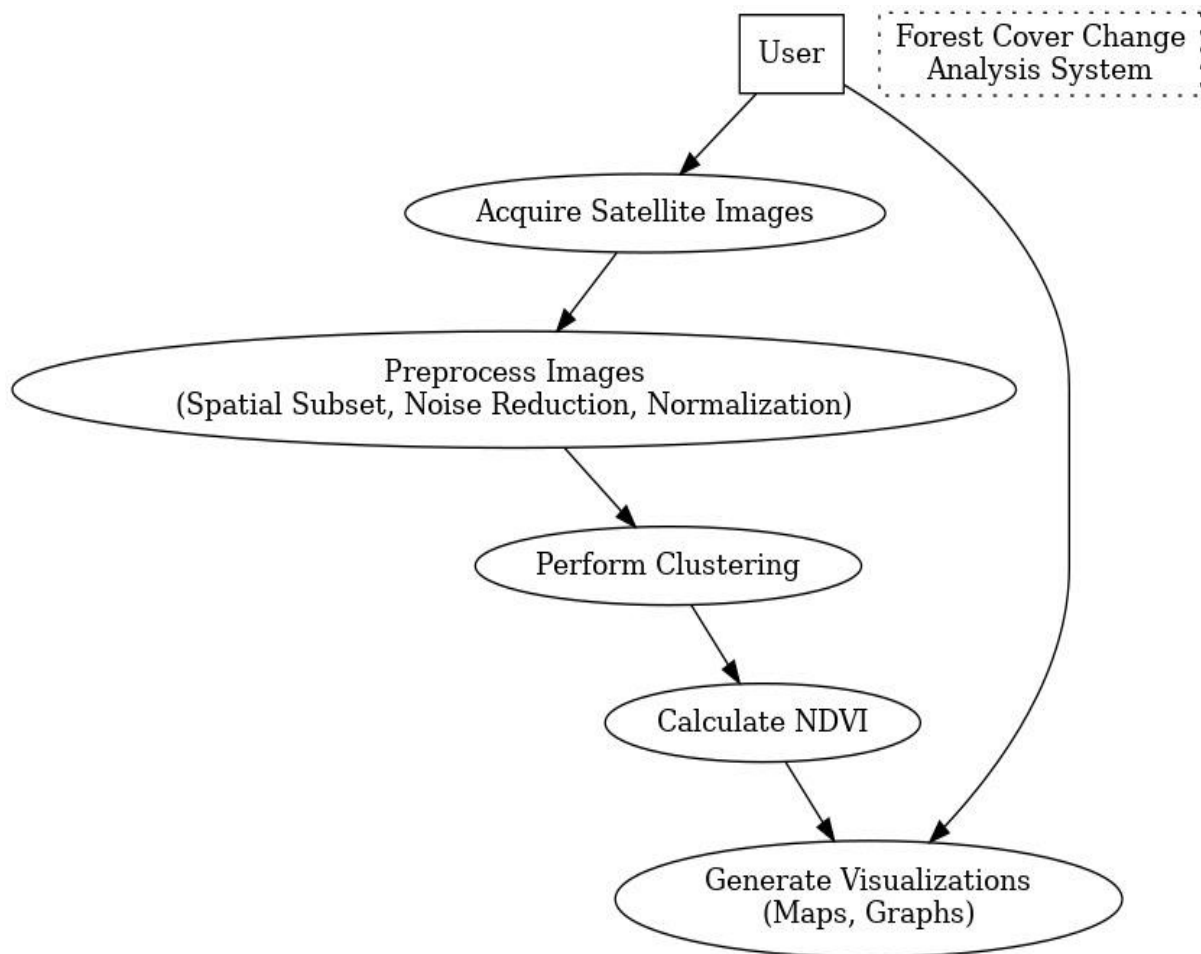
The project's design integrates a systematic workflow to ensure comprehensive analysis. It begins with the acquisition of satellite images, followed by preprocessing to enhance image quality and standardize data. Advanced algorithms are applied for clustering and spectral analysis, culminating in visualizations that highlight forest cover changes. The design ensures that each step builds upon the previous one, enabling robust and accurate results.

BASIC WORKFLOW:

- **Image Acquisition:** Collect high-resolution satellite images from ResourceSat and Cartosat via government portals like Bhuvan and Bhoonidhi. These portals ensure access to reliable and updated datasets.
- **Preprocessing:** Perform essential steps to prepare images for analysis:
 1. Spatial subset selection to focus on the study area.
 2. Noise reduction to remove artifacts and improve image clarity.
 3. Normalization to standardize data for consistent results.
- **Clustering:** Employ clustering algorithms to classify the region into categories based on vegetation density. The clusters provide insights into areas with dense forests, sparse vegetation, and urban developments.
- **Spectroscopy Analysis:** Compute NDVI values to identify healthy vegetation and degraded areas. NDVI serves as a reliable metric for understanding vegetation health over time.

- **Visualization:** Create maps, graphs, and comparative charts to illustrate forest cover changes over time. These visualizations provide a clear and intuitive representation of deforestation trends and hotspots.

USE CASE DIAGRAM



TEST RESULTS

RESULTS

- Satellite images from 2014 to 2024 were successfully analyzed, revealing substantial changes in forest cover.
- Results indicated a significant reduction in forested areas, particularly in suburban and peri-urban regions where urban expansion has been most pronounced.
- NDVI values demonstrated declining vegetation health in urbanized zones, with some regions showing near-complete deforestation.
- Visualization included:
 - Time-series maps that illustrated forest cover changes over the decade.

DEMONSTRATION

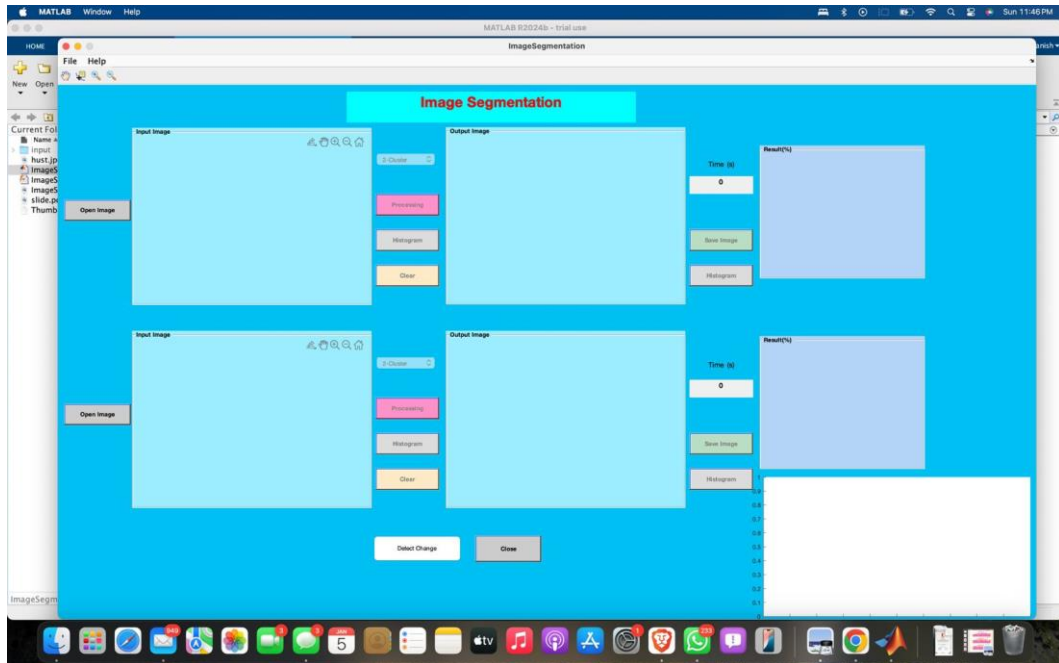
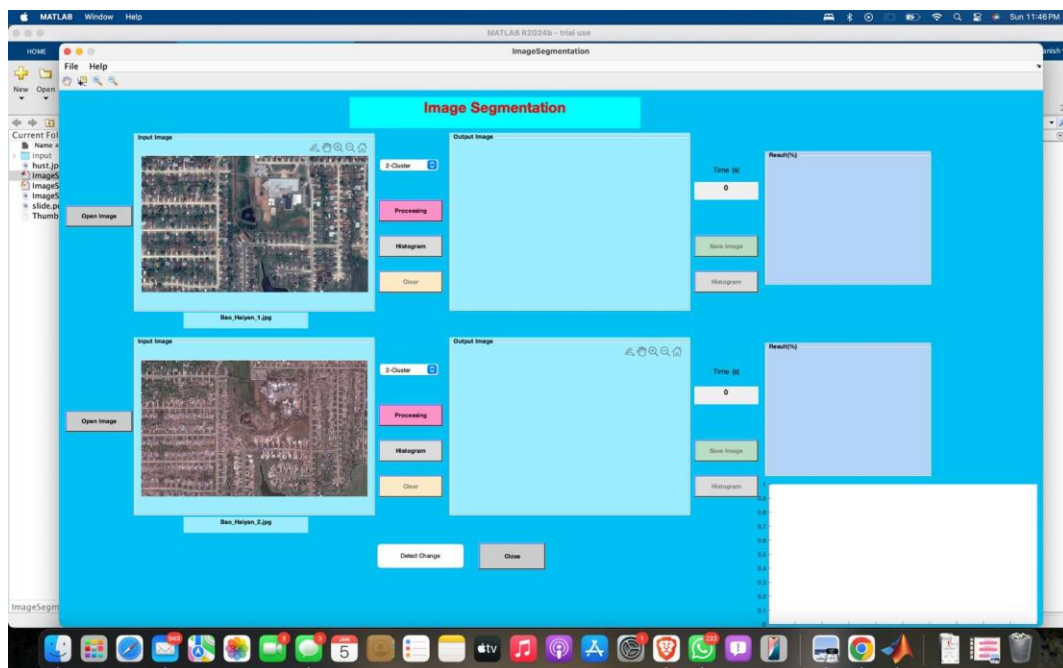
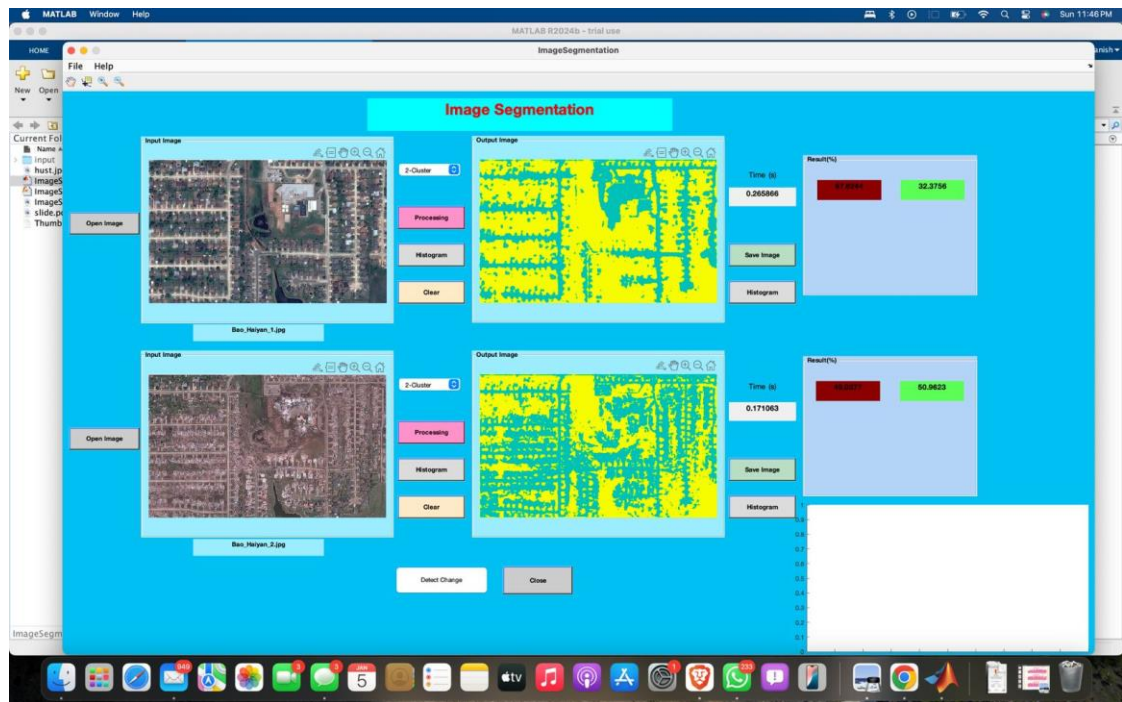


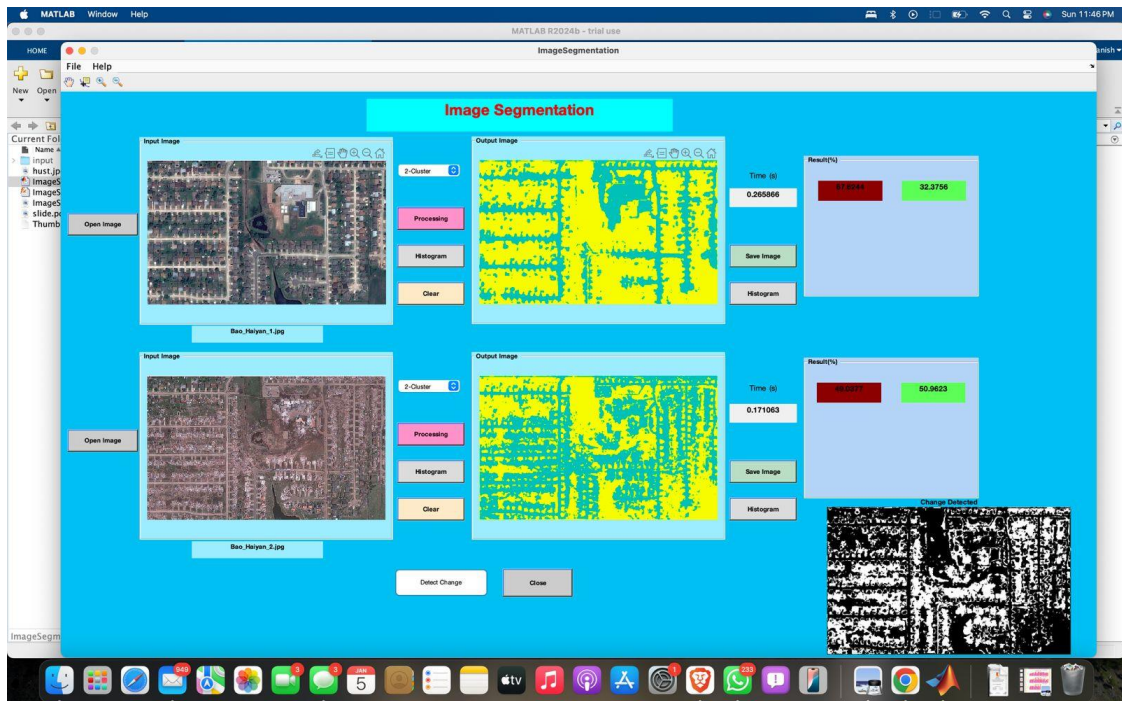
Image Processing Application



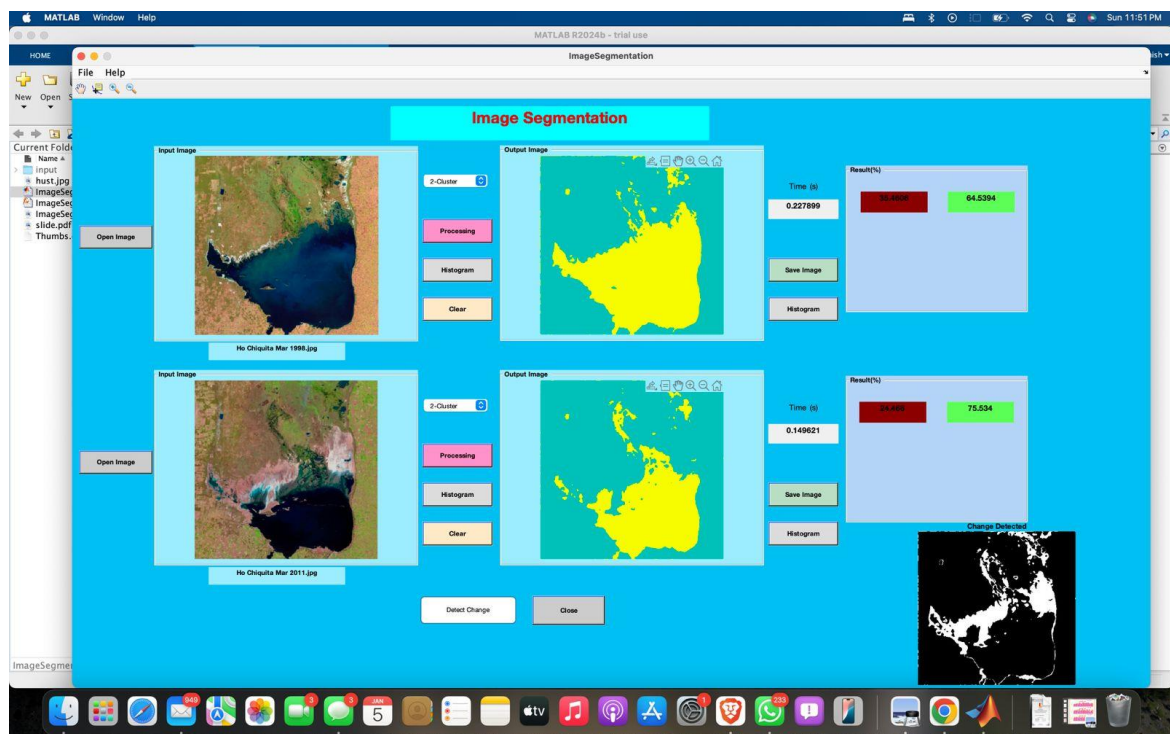
Uploading the Images



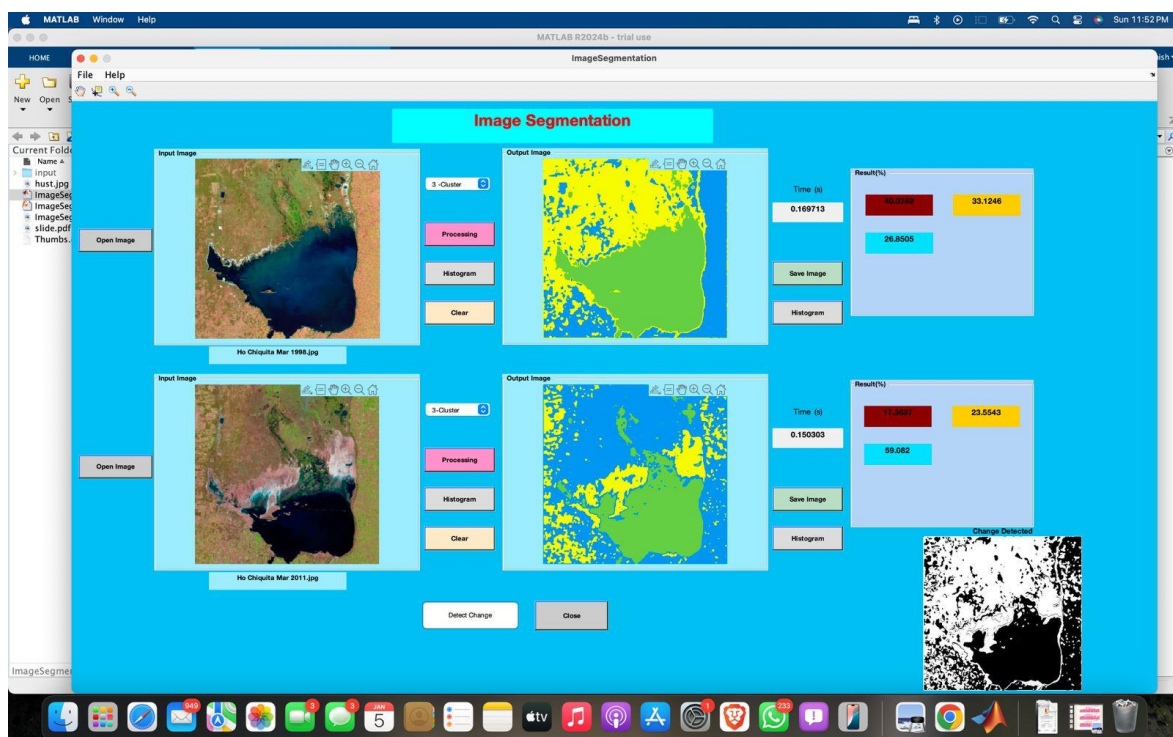
Output



Change Detected



2 Cluster Segmentation



3 Cluster Segmentation

CONCLUSION AND FUTURE SCOPE

CONCLUSION

The project effectively utilized satellite imagery and advanced image processing techniques to analyze forest cover changes in Delhi over the last decade. The findings underscored the critical need for sustainable urban planning and proactive measures to mitigate environmental degradation. By providing a scalable and reproducible workflow, this project contributes to the growing body of research on urban environmental management and climate change mitigation.

FUTURE SCOPE:

- Expanding the study to include real-time satellite data for dynamic monitoring of forest changes and urban expansion.
- Integrating machine learning models to enhance prediction accuracy and automate feature extraction, thereby reducing manual intervention.
- Developing a mobile or web-based platform to make analysis accessible to policymakers, researchers, and environmentalists. Such platforms could provide interactive dashboards, live updates, and predictive modeling features.
- Incorporating additional spectral indices, such as Enhanced Vegetation Index (EVI) and Soil-Adjusted Vegetation Index (SAVI), to provide a more nuanced understanding of vegetation health and land cover changes.
- Extending the analysis to assess the impact of deforestation on biodiversity, air quality, and water resources, enabling a holistic approach to urban environmental management.

- Studying the economic implications of deforestation and exploring cost-effective strategies for ecological restoration.
- Leveraging Artificial Intelligence to simulate future deforestation trends based on current urbanization rates.
- Exploring the role of public-private partnerships in afforestation projects and ecological preservation.

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